



PACKAGE INSERT
Product Name: Seacross Irinotecan Hydrochloride 20mg/ml Concentrate for Solution for Infusion
Name and Strength of Active Ingredient(s): IRINOTECAN HYDROCHLORIDE TRIHYDRATE 20mg/ml

Product Description
The product Irinotecan sol.iv.inf. 20mg/ml (40/100/300/500 mg/vial) is a sterile, clear yellow concentrated solution.
The sterile concentrate has been formulated as a 20mg/ml solution for infusion, supplied in amber Type I conventional glass vials at the presentations of 40mg/2ml, 100mg/5ml, 300mg/15ml and 500mg/25ml respectively.
To dilute with 5% glucose solution to a final concentration range of 0.12 to 2.8 mg/mL and 0.9% Sodium Chloride Injection to a final concentration range of 2.8 mg/mL.
Colourless or pale yellow clear solution and free from visible particles after dilution/reconstitution.

Pharmacodynamics
Irinotecan is a semisynthetic derivative of irinotecanthein. It is an antineoplastic agent which acts as a specific inhibitor of DNA topoisomerase I. It is metabolised by carboxylesterase in most tissues to SN-38, which was found to be more active than irinotecan in purified topoisomerase I and more cytotoxic than irinotecan against several murine and human tumour cell lines. The inhibition of DNA topoisomerase I by irinotecan or SN-38 induces single-strand DNA lesions which blocks the DNA replication fork and are responsible for the cytotoxicity. This cytotoxic activity was found time-dependent and was specific to the S phase.
In vitro, irinotecan and SN-38 were not found to be significantly recognised by the P-glycoprotein ^{MDR1} and displays cytotoxic activities against doxorubicin- and vinblastine-resistant cell lines.
Furthermore, irinotecan has a broad antitumour activity in vivo against murine tumour models (P03 pancreatic ductal adenocarcinoma, MA16/C mammary adenocarcinoma, C38 and C51 colon adenocarcinomas) and against human xenografts (C2-4 colon adenocarcinoma, Mx1 mammary adenocarcinoma, ST-15 and SC-16 gastric adenocarcinomas). Irinotecan is also active against tumours expressing the P-glycoprotein ^{MDR1} (vincristine- and doxorubicin-resistant P388 leukaemias).
Beside the antitumour activity of Irinotecan, the most relevant pharmacological effect of irinotecan is the inhibition of acetylcholinesterase.

Pharmacokinetics
With a dosage regimen of a 30-minute intravenous infusion of 100 to 750 mg/m² every three weeks, irinotecan showed a biphasic or triphasic elimination profile. The mean plasma clearance was 15 L/h/m² and the volume of distribution at steady state (V_{ss}): 157 L/m². The mean plasma half-life of the first phase of the triphasic model was 12 minutes, of the second phase 2.5 hours, and the terminal phase half-life was 14.2 hours. SN-38 showed a biphasic elimination profile with a mean terminal elimination half-life of 13.8 hours. At the end of the infusion, at the recommended dose of 350 mg/m², the mean peak plasma concentrations of irinotecan and SN-38 were 7.7 µg/ml and 56 ng/ml, respectively, and the mean area under the curve (AUC) values were 34 µg·h/ml and 451 ng·h/ml, respectively. A large interindividual variability in pharmacokinetic parameters is generally observed for SN-38.
Irinotecan and SN-38 exposure increase proportionally with irinotecan administered dose; their pharmacokinetics are independent of the number of previous cycles and of the administration schedule. Plasma protein binding for irinotecan and SN-38 was approximately 65% and 95% respectively.
More than 50% of an intravenously administered dose of irinotecan is excreted as unchanged drug, with 33% in the faeces mainly via the bile and 22% in urine.
Two metabolic pathways account each for at least 12% of the dose:
• Hydrolysis by carboxylesterase into active metabolite SN-38. SN-38 is mainly eliminated by glucuronidation, and further by biliary and renal excretion (less than 0.5% of the irinotecan dose) The SN-38 glucuronite is subsequently probably hydrolysed in the intestine.
• Cytochrome P450 3A4 enzymes-dependent oxidations resulting in opening of the outer piperidine ring with formation of APC (aminopentanoic acid derivate) and NPC (primary amine derivate) (see section Interaction with other medicaments).
Unchanged irinotecan is the major entity in plasma, followed by APC, SN-38 glucuronide and SN-38. Only SN-38 has significant cytotoxic activity.
Irinotecan clearance is decreased by about 40% in patients with bilirubinemia between 1.5 and 3 times the upper normal limit. In these patients a 200 mg/m² irinotecan dose leads to plasma drug exposure comparable to that observed at 350 mg/m² in cancer patients with normal liver parameters.

Indication
Treatment of patients with advanced colorectal cancer: In combination with 5-fluorouracil and folinic acid in patients without prior chemotherapy for advanced disease. As a single agent in patients who have failed an established 5-fluorouracil-containing treatment regimen.

Recommended Dose
Strictly follow the recommended dosage unless directed otherwise by the physician. All doses of irinotecan should be administered as an intravenous infusion over 30 to 90 minutes.

Single-Agent Dosage Schedules
Single-agent dosage schedules have been extensively studied for metastatic colorectal cancer.

Starting Dose
Weekly Dosage Schedule. The recommended single-agent starting dose of irinotecan is 125mg/m². A lower starting dose may be considered (e.g., 100mg/m²) for patients with any of the following conditions: prior extensive radiotherapy, performance status of 2, increased bilirubin levels, or gastric cancer. Treatment should be given in repeated 6-week cycles, comprising weekly treatment for 4 weeks, followed by a 2-week rest. Once-Every-2-Week Dosage Schedule. The usual recommended starting dose of irinotecan is 250mg/m² every 2 weeks by intravenous infusion. A lower starting dose may be considered (e.g., 200mg/m²) for patients with any of the following conditions: age 65 years and older, prior extensive radiotherapy, performance status of 2, increased bilirubin levels, or gastric cancer.
Once-Every-3-Week Dosage Schedule. The usual recommended starting dose of irinotecan for the once-3-week dosage schedule is 350mg/m². A lower starting dose may be considered (e.g., 300mg/m²) for patients with any of the following conditions: age 65 years and older, prior extensive radiotherapy, performance status of 2, increased bilirubin levels, or gastric cancer.

Special population
Elderly
The dose should be chosen carefully in this population due to their greater frequency of decreased biological functions. This population should require more intensive surveillance (See section **Pharmacokinetic Properties-Pharmacokinetics in Special Populations**).

Patients with Impaired Hepatic Function
In patients with hepatic dysfunction, the following starting doses are recommended:

Table 1: Starting Doses in Patients with Hepatic Dysfunction: Single-Agent Weekly Regimen		
Serum Total Bilirubin Concentration	Serum ALT/AST Concentration	Starting Dose, mg/m ²
1.5-3.0 x IULN	≤5.0 x IULN	60
3.1-5.0 x IULN	≤5.0 x IULN	50
<1.5 x IULN	5.1-20.0 x IULN	60
1.5-5.0 x IULN	5.1-20.0 x IULN	40

Table 2: Starting Doses in Patients with Hepatic Dysfunction: Single-Agent Once-Every-3-Week Regimen

Serum Total Bilirubin Concentration	Starting Dose, mg/m ²
1.5-3.0 x IULN	200
>3.0 x IULN	Not Recommended*

*The safety and pharmacokinetics of irinotecan given once-every-3-weeks have not been defined in patients with bilirubin >3.0 x institutional upper limit of normal (IULN) and this schedule cannot be recommended in these patients.

Patients with Impaired Renal Function
Studies in this population have not been conducted (See section **Pharmacokinetic Properties-Pharmacokinetics in Special Populations**). Therefore, caution should be undertaken in patients with impaired renal function. Irinotecan is not recommended for use in patients on dialysis.

Combination-Agent Dosage Schedules
Starting Dose
Irinotecan in Combination with 5-Fluorouracil (5-FU) and folinic acid (FA). Irinotecan in combination with 5-FU and folinic acid is recommended for use in patients with metastatic colorectal cancer. For all regimens, the dose of folinic acid should be administered immediately after irinotecan, with the administration of 5-FU to occur immediately after receipt of folinic acid. The currently recommended regimens are shown below:

Regimen 1 (6-week cycle with bolus 5-FU/FA): The recommended starting dose is 125 mg/m² of irinotecan, 500 mg/m² bolus 5-FU, and 20 mg/m² bolus folinic acid.
Regimen 2 (6-week cycle with infusional 5-FU/FA): The recommended starting dose is 180 mg/m² of irinotecan, 400 mg/m² bolus 5-FU, 600 mg/m² 5-FU infusion, and 200 mg/m² folinic acid.
Lower starting doses may be considered for irinotecan (e.g., 100mg/m²) and 5-FU (e.g., 400mg/m²) for patients with any of the following conditions: age 65 years and older, prior extensive radiotherapy, performance status of 2, increased bilirubin levels, or gastric cancer. Treatment should be given in repeated 6-week cycles, comprising weekly treatment for 4 weeks, followed by a 2-week rest.

Duration of Treatment
For both single-agent and combination-agent regimens, treatment with additional cycles of irinotecan may be continued indefinitely in patients who attain a partial response or in patients whose cancer remains stable. Patients should be carefully monitored for toxicity and should be removed from therapy if unacceptable toxicity occurs that is not responsive to dose modification and routine supportive care.

Dose Modification Recommendations
The recommended dose modifications during a cycle of therapy and at the start of each subsequent cycle of therapy for single-agent dosage schedules are described in **Table 3**. These recommendations are based on toxicities commonly observed with the administration of irinotecan. For modifications at the start of a subsequent cycle of therapy, the dose of irinotecan should be decreased relative to the initial dose of the previous cycle.
The recommended dose modifications during a cycle of therapy and at the start of each subsequent cycle of therapy for irinotecan, 5-FU, and folinic acid are described in **Table 4**.
All dose modifications should be based on the worst preceding toxicity. A new cycle of therapy should not begin until the toxicity has recovered to grade 2 or less. Treatment may be delayed 1 to 2 weeks to allow for recovery from treatment-related toxicity. If the patient has not recovered, consideration should be given to discontinuing irinotecan.

Table 3: RECOMMENDED DOSE MODIFICATIONS FOR SINGLE-AGENT SCHEDULE
A new cycle of therapy should not begin until the granulocyte count has recovered to ≥1500/mm³, and the platelet count has recovered to ≥100,000/mm³, and treatment-related diarrhea is fully resolved. Treatment should be delayed 1 to 2 weeks to allow for recovery from treatment-related toxicities. If the patient has not recovered after a 2-week delay, consideration should be given to discontinuing irinotecan.

Toxicity NCI Grade ^a (Value)	During a Cycle of Therapy		At the Start of the Next Cycle of Therapy (After Adequate Recovery), Compared with the Starting Dose in the Previous Cycle ^b	
	Weekly	Weekly	Once Every 2 or 3 Week	
No toxicity	Maintain dose level	↓ 25mg/m ² up to a maximum dose of 150mg/m ²	Maintain dose level	
Neutropenia				
1 (1500 to 1999/mm ³)	Maintain dose level	Maintain dose level	Maintain dose level	
2 (1000 to 1499/mm ³)	↓ 25mg/m ²	Maintain dose level	Maintain dose level	
3 (500 to 999/mm ³)	Omit dose, then ↓ 25mg/m ² when resolved to ≤grade 2	↓ 25mg/m ²	↓ 50mg/m ²	
4 (<500/mm ³)	Omit dose, then ↓ 50mg/m ² when resolved to ≤grade 2	↓ 50mg/m ²	↓ 50mg/m ²	
Neutropenic fever (grade 4 neutropenia & ≥ grade 2 fever)	Omit dose, then ↓ 50mg/m ² when resolved	↓ 50mg/m ²	↓ 50mg/m ²	
Other hematologic toxicities	Dose modifications for leukopenia, thrombocytopenia, and anemia during a cycle of therapy and at the start of subsequent cycles of therapy are also based on NCI toxicity criteria and are the same as recommended for neutropenia above.			
Diarrhea 1 (2-3 stools/day > pretx ^c) 2 (4-6 stools/day > pretx ^c) 3 (7-9 stools/day > pretx ^c) 4 (≥10 stools/day > pretx ^c)	Maintain dose level ↓ 25mg/m ² Omit dose, then ↓ 25mg/m ² when resolved to ≤grade 2	Maintain dose level ↓ 25mg/m ² ↓ 50mg/m ²	Maintain dose level ↓ 50mg/m ² ↓ 50mg/m ²	
Other non-hematologic toxicities ^d 1 2 3 4	Maintain dose level ↓ 25 mg/m ² Omit dose, then ↓ 25 mg/m ² when resolved to ≤grade 2	Maintain dose level ↓ 25 mg/m ² ↓ 25 mg/m ²	Maintain dose level ↓ 50 mg/m ² ↓ 50 mg/m ²	

^aAll dose modifications should be based on the worst preceding toxicity.
^bNational Cancer Institute Common Toxicity Criteria.
^cPre-treatment
^dExcludes alopecia, anorexia, asthenia.

Table 4: Recommended Dose Modifications for Irinotecan/5-Fluorouracil/Folinic Acid Combination Schedules

Patients should return to pre-treatment bowel function without requiring antidiarrhea medications for at least 24 hours before the next chemotherapy administration. A new cycle of therapy should not begin until the granulocyte count has recovered to ≥1500/mm³, and the platelet count has recovered to ≥100,000/mm³, and treatment-related diarrhea is fully resolved. Treatment should be delayed 1 to 2 weeks to allow for recovery from treatment-related toxicities. If the patient has not recovered after a 2-week delay, consideration should be given to discontinuing irinotecan.

Grade ^a (Value)	During a Cycle of Therapy	At the Start of Subsequent Cycles of Therapy
No toxicity	Maintain dose level	Maintain dose level
Neutropenia		
1 (1500 to 1999/mm ³)	Maintain dose level ^b	Maintain dose level ^b
2 (1000 to 1499/mm ³)	↓ 1 dose level ^b	Maintain dose level
3 (500 to 999/mm ³)	Omit dose, then ↓ 1 dose level when resolved to ≤Grade 2 ^c	↓ 1 dose level ^b
4 (< 500/mm ³)	Omit dose, then ↓ 2 dose levels when resolved to ≤Grade 2 ^c	↓ 2 dose levels
Neutropenic fever(Grade 4 neutropenia & ≥Grade 2 fever)	Omit dose, then ↓ 2 dose levels when resolved	↓ 2 dose levels
Other hematologic toxicities	Dose modifications for leukopenia or thrombocytopenia during a cycle of therapy and at the start of subsequent cycles of therapy are also based on NCI toxicity criteria and are the same as recommended for neutropenia above.	
Diarrhea 1 (2-3 stools/day > pretx ^c) 2 (4-6 stools/day > pretx ^c) 3 (7-9 stools/day > pretx ^c) 4 (≥10 stools/day > pretx ^c)	Delay dose until resolved to baseline (bsl), then give same dose Omit dose, then ↓ 1 dose level when resolved to bsl Omit dose, then ↓ 1 dose level when resolved to bsl Omit dose, then ↓ 2 dose levels when resolved to bsl	Maintain dose level Maintain dose level ↓ 1 dose level ↓ 2 dose levels
Other non-hematologic Toxicities ^d 1 2 3 4	Maintain dose level Omit dose, then ↓ 1 dose level when resolved to ≤Grade 1 Omit dose, then ↓ 1 dose level when resolved to ≤Grade 2 Omit dose, then ↓ 2 dose levels when resolved to ≤Grade 2 <i>For mucositis/stomatitis decrease only 5-FU, not irinotecan.^e</i>	Maintain dose level Maintain dose level ↓ 1 dose level ↓ 2 dose levels <i>For mucositis/stomatitis decrease only 5-FU, not irinotecan.^e</i>

^aDose modification refers to irinotecan and 5-FU; LV dose remains fixed at 20 mg/m² (not adjusted).
^bNational Cancer Institute Common Toxicity Criteria.
^cRefers to initial dose used in previous cycle.
^dIrinotecan: dose level reductions = 25 mg/m² decrements; 5-Fluorouracil: dose level reductions = 100 mg/m² decrements.
^ePre-treatment
^fExcludes alopecia, anorexia, asthenia.
^gFor mucositis/stomatitis decrease only 5-FU, not irinotecan.

Contraindication
• Chronic inflammatory bowel disease and/or bowel obstruction (see section warning and precautions).
• History of severe hypersensitivity reactions to irinotecan hydrochloride trihydrate or to one of the excipients of irinotecan.
• Lactation (see section warning and precautions and section pregnancy and lactation).
• Bilirubin > 3 times the upper limit of the normal range (see warning and precautions).
• Severe bone marrow failure.
• WHO performance status > 2.
• Concomitant use with St John's Wort (see section Interaction with other medicaments).

Warning and Precautions
The use of SEACROSS IRINOTECAN should be confined to units specialised in the administration of cytotoxic chemotherapy and it should only be administered under the supervision of a physician qualified in the use of anticancer chemotherapy.

Given the nature and incidence of adverse events, Irinotecan will only be prescribed in the following cases after the expected benefits have been weighed against the possible therapeutic risks:
• in patients presenting a risk factor, particularly those with a WHO performance status = 2.
• in the few rare instances where patients are deemed unlikely to observe recommendations regarding management of adverse events (need for immediate and prolonged antidiarrhoeal treatment combined with high fluid intake at onset of delayed diarrhea). Strict hospital supervision is recommended for such patients.

When Irinotecan is used in monotherapy, it is usually prescribed with the every-3-week-dosage schedule. However, the weekly-dosage schedule (see section Pharmacodynamics/pharmacokinetics) may be considered in patients who may need a closer follow-up or who are at particular risk of severe neutropenia.

Delayed diarrhea
Patients should be made aware of the risk of delayed diarrhea occurring more than 24 hours after the administration of Irinotecan and at any time before the next cycle. In monotherapy, the median time of onset of the first liquid stool was on day 5 after the infusion of Irinotecan. Patients should quickly inform their physician of its occurrence and start appropriate therapy immediately.
Patients with an increased risk of diarrhoea are those who had a previous abdominal/pelvic radiotherapy, those with baseline hyperleucocytosis, those with performance status ≥2 and women. If not properly treated, diarrhoea can be life-threatening, especially if the patient is concomitantly neutropenic. As soon as the first liquid stool occurs, the patient should start drinking large volumes of beverages containing electrolytes and an appropriate antidiarrhoeal therapy must be initiated immediately. This antidiarrhoeal treatment will be prescribed by the department where Irinotecan has been administered. After discharge from the hospital, the patients should obtain the prescribed drugs so that they can treat the diarrhoea as soon as it occurs. In addition, they must inform their physician or the department administering Irinotecan when/if diarrhoea is occurring.
The currently recommended antidiarrhoeal treatment consists of high doses of loperamide (4 mg for the first intake and then 2 mg every 2 hours). This therapy should continue for 12 hours after the last liquid stool and should not be modified. In no instance should loperamide be administered for more than 48 consecutive hours at these doses, because of the risk of paralytic ileus, nor for less than 12 hours. In addition to the anti-diarrhoeal treatment, a prophylactic broad spectrum antibiotic should be given, when diarrhoea is associated with severe neutropenia (neutrophil count < 500 cells/mm³).
In addition to the antibiotic treatment, hospitalisation is recommended for management of the diarrhoea, in the following cases:
- Diarrhoea associated with fever,
- Severe diarrhoea (requiring intravenous hydration),
- Diarrhoea persisting beyond 48 hours following the initiation of high-dose loperamide therapy.
Loperamide should not be given prophylactically, even in patients who experienced delayed diarrhoea at previous cycles.
In patients who experienced severe diarrhoea, a reduction in dose is recommended for subsequent cycles (see recommended dosage).

Haematology
Neutropenia has been significantly higher in patients who received previous pelvic/abdominal irradiation than in those who had not received such irradiation. Patients with baseline serum total bilirubin levels of 1.0 mg/dL or more have also had a significantly greater likelihood of neutropenia than those with bilirubin levels that were less than 1.0 mg/dL.
Weekly monitoring of complete blood cell counts is recommended during Irinotecan treatment. Patients should be aware of the risk of neutropenia and the significance of fever. Febrile neutropenia (temperature > 38° C and neutrophil count ≤1,000 cells/mm³) should be urgently treated in the hospital with broad-spectrum intravenous antibiotics.
In patients who experienced severe haematological events, a dose reduction is recommended for subsequent administration (see section Recommended dosage)
There is an increased risk of infections and haematological toxicity in patients with severe diarrhoea. In patients with severe diarrhoea, complete blood cell counts should be performed.

Liver impairment
Liver function tests should be performed at baseline and before each cycle.
Weekly monitoring of complete blood counts should be conducted in patients with bilirubin ranging from 1.5 to 3 times ULN, due to decrease of the clearance of irinotecan (see section Pharmacokinetic) and thus increasing the risk of hematotoxicity in this population. For patients with a bilirubin > 3 times ULN (see section Contraindications).

Nausea and vomiting
A prophylactic treatment with antiemetics is recommended before each treatment with Irinotecan. Nausea and vomiting have been frequently reported. Patients with vomiting associated with delayed diarrhoea should be hospitalised as soon as possible for treatment.

Acute cholinergic syndrome
If acute cholinergic syndrome appears (defined as early diarrhoea and various other signs and symptoms such as sweating, abdominal cramping, myosis and salivation), atropine sulphate (0.25 mg subcutaneously) should be administered unless clinically contraindicated (see section Adverse effect). These symptoms may be observed during or shortly after infusion of irinotecan, are thought to be related to the anticholinesterase activity of the irinotecan parent compound, and are expected to occur more frequently with higher irinotecan doses.
Caution should be exercised in patients with asthma. In patients who experienced an acute and severe cholinergic syndrome, the use of prophylactic atropine sulphate is recommended with subsequent doses of Irinotecan.

Respiratory disorders
Interstitial pulmonary disease presenting as pulmonary infiltrates is uncommon during irinotecan therapy. Interstitial pulmonary disease can be fatal. Risk factors possibly associated with the development of interstitial pulmonary disease include the use of pneumotoxic drugs, radiation therapy and colony stimulating factors. Patients with risk factors should be closely monitored for respiratory symptoms before and during irinotecan therapy.

Extravasation
While irinotecan is not a known vesicant, care should be taken to avoid extravasation and the infusion site should be monitored for signs of inflammation. Should extravasation occur, flushing the site and application of ice is recommended.

Elderly
Due to the greater frequency of decreased biological functions, in particular hepatic function, in elderly patients, dose selection with Irinotecan should be cautious in this population (see section recommended dosage).

Chronic inflammatory bowel disease and/or bowel obstruction
Patients must not be treated with Irinotecan until resolution of the bowel obstruction (see section Contraindication).

Renal function
Increases in serum creatinine or blood urea nitrogen have been observed. There have been cases of acute renal failure. These events have generally been attributed to complications of infection or to dehydration related to nausea, vomiting, or diarrhoea. Rare instances of renal dysfunction due to tumour lysis syndrome have also been reported.

Irradiation therapy
Patients who have previously received pelvic/abdominal irradiation are at increased risk of myelosuppression following the administration of irinotecan. Physicians should use caution in treating patients with extensive prior irradiation (e.g. >25% of bone marrow irradiated and within 6 weeks prior to start of treatment with irinotecan). Dosing adjustment may apply to this population (see section Recommended dosage).

Cardiac disorders
Myocardial ischaemic events have been observed following irinotecan therapy predominately in patients with underlying cardiac disease, other known risk factors for cardiac disease, or previous cytotoxic chemotherapy (see section Adverse effects).
Consequently, patients with known risk factors should be closely monitored, and action should be taken to try to minimize all modifiable risk factors (e.g. smoking, hypertension, and hyperlipidaemia).

Vascular disorders
Irinotecan has been rarely associated with thromboembolic events (pulmonary embolism, venous thrombosis, and arterial thromboembolism) in patients presenting with multiple risk factors in addition to the underlying neoplasm.

Immunosuppressant effects/increased susceptibility to infections
Administration of live or live-attenuated vaccines in patients immunocompromised by chemotherapeutic agents including irinotecan, may result in serious or fatal infections. Vaccination with a live vaccine should be avoided in patients receiving irinotecan. Killed or inactivated vaccines may be administered; however, the response to such vaccines may be diminished.

Others
Since this medicinal contains sorbitol, it is unsuitable in hereditary fructose intolerance.
Infrequent cases of renal insufficiency, hypotension or circulatory failure have been observed in patients who experienced episodes of dehydration associated with diarrhoea and/or vomiting, or sepsis.
Contraceptive measures must be taken during and for at least three months after cessation of therapy. Concomitant administration of irinotecan with a strong inhibitor (e.g. ketoconazole) or inducer (e.g. rifampicin, carbamazepine, phenobarbital, phenytoin, St John's Wort) of CYP3A4 may alter the metabolism of irinotecan and should be avoided (see section Interaction with other medicaments).

Interactions With Other Medicaments
Interaction between irinotecan and neuromuscular blocking agents cannot be ruled out. Since Irinotecan has anticholinesterase activity, drugs with anticholinesterase activity may prolong the neuromuscular blocking effects of suxamethonium and the neuromuscular blockade of non-depolarising drugs may be antagonised. Concomitant administration of CYP3A4-inducing anticonvulsant drugs (e.g., carbamazepine, phenobarbital or phenytoin) leads to reduced exposure to irinotecan, SN-38 and SN-38 glucuronide and reduced pharmacodynamic effects. The effects of such anticonvulsant drugs was reflected by a decrease in AUC of SN-38 and SN-38G. In addition to induction of cytochrome P450 3A4 enzymes, enhanced glucuronidation and enhanced biliary excretion may play a role in reducing exposure to irinotecan and its metabolites. Co-administration of ketoconazole resulted in a decrease in the AUC of APC and in an increase in the AUC of SN-38.
Caution should be exercised in patients concurrently taking drugs known to inhibit (e.g., ketoconazole) or induce (e.g., rifampicin, carbamazepine, phenobarbital or phenytoin) drug metabolism by cytochrome P450 3A4. Concurrent administration of irinotecan with an inhibitor/inducer of this metabolic pathway may alter the metabolism of irinotecan and should be avoided (see section Warning and precautions).
St. John's Wort decreases SN-38 plasma levels. As a result, St. John's Wort should not be administered with irinotecan (see section Contraindication).
Co-administration of 5-fluorouracil/folinic acid in the combination regimen does not change the pharmacokinetics of irinotecan.
Atazanavir sulphate. Co-administration of atazanavir sulfate, a CYP3A4 and UGT1A1 inhibitor, has the potential to increase systemic exposure to SN-38, the active metabolite of irinotecan. Physicians should take this into consideration when co-administering these drugs.

Interactions common to all cytotoxics:
The use of antineoplastic agents is common due to increased risk of thrombotic events in tumoral diseases. If vitamin K antagonist anticoagulants are indicated, an increased frequency in the monitoring of INR (International Normalised Ratio) is required due to their narrow therapeutic index, the high intra-individual variability of blood thrombogenicity and the possibility of interaction between oral anticoagulants and anticancer chemotherapy.

Concomitant use contraindicated:
- Yellow fever vaccine: risk of fatal generalised reaction to vaccines

Concomitant use not recommended:
- Live attenuated vaccines (except yellow fever): risk of systemic, possible fatal disease (eg-infections). This risk is increased in subjects who are already immunosuppressed by their underlying disease.
- Use an inactivated vaccine where this exists (poliomyelitis)
- Phenytoin: Risk of exacerbation of convulsions resulting from the decrease of phenytoin digestive absorption by cytotoxic drug or risk of toxicity enhancement due to increased hepatic metabolism by phenytoin

Concomitant use to take into consideration:
- Ciclosporine, Tacrolimus: Excessive immunosuppression with risk of lymphoproliferation
There is no evidence that the safety profile of irinotecan is influenced by cetuximab or vice versa.

Pregnancy and Lactation
Pregnancy
There are no adequate and well-controlled studies of irinotecan in pregnant women. Irinotecan is teratogenic in rats and rabbits. Irinotecan may cause fetal harm when administered to a pregnant woman. Women of childbearing potential should not be started on irinotecan until pregnancy is excluded. Pregnancy should be avoided if either partner is receiving irinotecan. Due to the potential for genotoxicity, advise female patients of reproductive potential to use highly effective contraception during treatment and for 6 months after the last dose of irinotecan. Due to the potential for genotoxicity, advise male patients with female partners of reproductive potential to use effective contraception during treatment and for 3 months after the last dose of irinotecan.

Lactation
The available data are limited to one patient only. Irinotecan and its active metabolite SN-38 were measured in the milk of one lactating patient. The effect on newborn/infants is unknown. Because of the potential for serious adverse reactions in nursing infants, it is recommended not to breastfeed when receiving therapy with irinotecan. In rats, radioactively appeared in the milk within 5 minutes of intravenous administration of radiolabelled irinotecan and was concentrated up to 65-fold at 4 hours after administration relative to plasma concentrations.

Side Effects
The most common, dose-limiting adverse reactions of irinotecan are delayed diarrhoea (occurring more than 24 hours after administration) and blood disorders including neutropenia, anaemia and thrombocytopenia.
Neutropenia is a dose-limiting toxic effect. Neutropenia was reversible and not cumulative; the median day to nadir was 8 days whatever the use in monotherapy or in combination therapy.
Very commonly severe transient acute cholinergic syndrome was observed.

The main symptoms were defined as early diarrhoea and various other symptoms such as abdominal pain, sweating, myosis and increased salivation occurring during or within the first 24 hours after the infusion of irinotecan. These symptoms disappear after atropine administration (see section Warning and precautions).
MONOTHERAPY
The following adverse reactions considered to be possibly or probably related to the administration of irinotecan at the recommended dose of 350 mg/m² in monotherapy. Within each frequency grouping, adverse reactions are presented in order of decreasing seriousness. Frequencies are defined as: very common, common, uncommon, rare, and very rare.

Adverse Reactions Reported with Irinotecan in Monotherapy (350 mg/m ² every 3 weeks schedule)		
MedDRA System Organ Class	Frequency Category	Preferred Term
Infections and infestations	Common	Infection
Blood and lymphatic system disorders	Very common	Neutropenia
	Very common	Anaemia
	Common	Thrombocytopenia
	Common	Febrile neutropenia
Metabolism and nutrition disorders	Very common	Decreased appetite
Nervous system disorders	Very common	Cholinergic syndrome
Gastrointestinal disorders	Very common	Diarrhoea
	Very common	Vomiting
	Very common	Nausea
	Very common	Abdominal pain
	Common	Constipation
Skin and subcutaneous tissue disorders	Very common	Alopecia (reversible)
General disorders and administration site conditions	Very common	Mucosal inflammation
	Very common	Pyrexia
	Very common	Asthenia
Investigations	Common	Blood creatinine increased
	Common	Transaminases (SGPT and SGOT) increased
	Common	Bilirubin increased
	Common	Blood alkaline phosphatase increased

COMBINATION THERAPY
Adverse reactions detailed in this section refer to irinotecan.
There is no evidence that the safety profile of irinotecan is influenced by cetuximab or vice versa. In combination with cetuximab, additional reported adverse reactions were those expected with cetuximab (such as acneiform rash). For information on adverse reactions on irinotecan in combination with cetuximab, also refer to their respective summary of product characteristics.

Adverse drug reactions reported in patients treated with capecitabine in combination with irinotecan in addition to those seen with capecitabine monotherapy or seen at a higher frequency grouping compared to capecitabine monotherapy include: *Very common, all grade adverse drug reactions:* thrombosis/embolism; *Common, all grade adverse drug reactions:* hypersensitivity reaction, cardiac ischemia/infarction; *Common, grade 3 and grade 4 adverse drug reactions:* febrile neutropenia. For complete information on adverse reactions of capecitabine, refer to the capecitabine summary product of characteristics.

Grade 3 and Grade 4 adverse drug reactions reported in patients treated with capecitabine in combination with irinotecan and bevacizumab in addition to those seen with capecitabine monotherapy or seen at a higher frequency grouping compared to capecitabine monotherapy include: *Common, grade 3 and grade 4 adverse drug reactions:* neutropenia, thrombosis/embolism, hypertension, and cardiac ischemia/infarction. For complete information on adverse reactions of capecitabine and bevacizumab, refer to the respective capecitabine and bevacizumab summary of product characteristics.

Artwork Panel	Artwork Ref: Irinotecan-PIL-MA	Internal Code:47470201800102	Artwork Code:
	Version No: 02	Date: 26/07-2023	Colour References: PMS1 K100 ■
Product: Irinotecan hydrochloride	Pack Type: PIL	Substrate Weight: 60gsm	如果非汇宇场地包装 说明书印刷时需要确认是否折叠 HUIYU by Design
Client: Medispec (M) Sdn Bhd	Dimensions: 240x480mm	Fonts: Arial,Arial-bold, Min Font Point:6	